



Grade 11 Geography

3. Hilly Landscapes & Basaltic Plateaus

Learning Guide

1. Overview of Landscape Types

- **Hilly Landscapes**
- **Basaltic Plateaux**
- **Processes:**
 - *Chemical Weathering*
 - *Mechanical Weathering*
 - *Erosion*
- **Key Concepts:**
 - *Scarp Retreat*
 - *Back Wasting*

2. Hilly Landscapes – Characteristics

- Found in **humid and arid regions**
- Rocks are **horizontal, sedimentary, and resistant to erosion**
- **Slopes:**
 - Rounded in **humid regions**
 - Angular and steep in **arid regions**
- **Soil:**
 - Deep in humid areas
 - Thin and coarse in arid areas

3. Processes in Humid Areas (Chemical Weathering)

- Dominates due to **high moisture and temperature**
- **Hydrolysis:** $\text{CO}_2 + \text{rainwater} \rightarrow \text{carbonic acid}$
- **Dissolution:** Materials dissolve in water
- **Oxidation:** Iron + oxygen $\rightarrow$ rust
- **Biological activity:** Roots & bacteria weaken rocks
- **Animal activity:** Burrowing exposes rock

4. Processes in Arid Areas (Mechanical Weathering)

- Dominates due to **low water availability**
- No chemical change $\rightarrow$ **physical breakdown only**
- **Ice wedging:** Freezing water expands cracks
- **Exfoliation:** Heating & cooling $\rightarrow$ expansion/contraction
- **Sandblasting:** Wind-driven sand erodes rock
- **Lightning:** Splits rocks with heat

5. Basaltic Plateaux – Characteristics

- **Flat-topped and elevated landforms**
- Formed by **lava flows (fissures) + crust uplift**
- Found across **continents**
- Have **steep sides and vertical joints**

-  Example: **Drakensberg Mountains**



6. Formation of Basaltic Plateaux

- Formed during **breakup of Gondwana** (~170 million years ago)
- Lava flowed through **large fissures**
- Cooled into **horizontal basalt layers**
- Resulted in:
 - **Steep cliffs**
 - **Deep joints**
- Lava spread as far as **Zimbabwe**
- Present landscape: **Black basalt cliffs (Drakensberg)**



7. Scarp Retreat

- **Definition:** Backward movement of a **steep slope (scarp)**
- **Cause:**
 - **Weathering and erosion at the base**
- **Key Idea:**
 - Slope **keeps its shape and height**
 - **BUT moves backward**, reducing width
- **Result:**
 - Landscape shifts backward over time
 - Leaves behind **flatter surfaces (pediplains)**



8. Back Wasting

- **Definition:** Gradual **wearing away of the entire slope**
- **Process:**
 - Weathering acts on the **whole slope surface**
- **Result:**
 - Slope becomes **less steep over time**
 - Leads to **broad valleys and gentle landforms**



9. Key Summary

- **Humid areas → Chemical weathering → Rounded hills**
- **Arid areas → Mechanical weathering → Rugged terrain**
- **Basaltic plateaux → Formed by lava flows**
- **Scarp retreat + Back wasting → Shape landscapes over time**
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